
Decoding Identity Threat Detection and Response (ITDR)

What is Identity Threat Detection & Response (ITDR)?

Identity Threat Detection and Response (ITDR) is a vertical of security which involves identifying, reducing, and responding to identity-based threats, including significant threats like identity attacks that ITDR addresses.

This includes compromised user accounts, fraudulent access, and more. Compromised credentials pose a significant security risk by allowing unauthorized access that mimics legitimate user activity.

Gartner defines ITDR as ***“a security discipline that encompasses threat intelligence, best practices, a knowledge base, tools, and processes to protect identity systems. It works by implementing detection mechanisms, investigating suspicious posture changes and activities, and responding to attacks to restore the integrity of the identity infrastructure.”***

With multiple users accessing company resources from multiple devices, it has become important to come up with a security strategy which detects and responds to threats on an identity-level. Hence, ITDR solutions cannot be called a product, it is a set of procedures which creates a security framework. We will dive deeper into these procedures later in this blog.

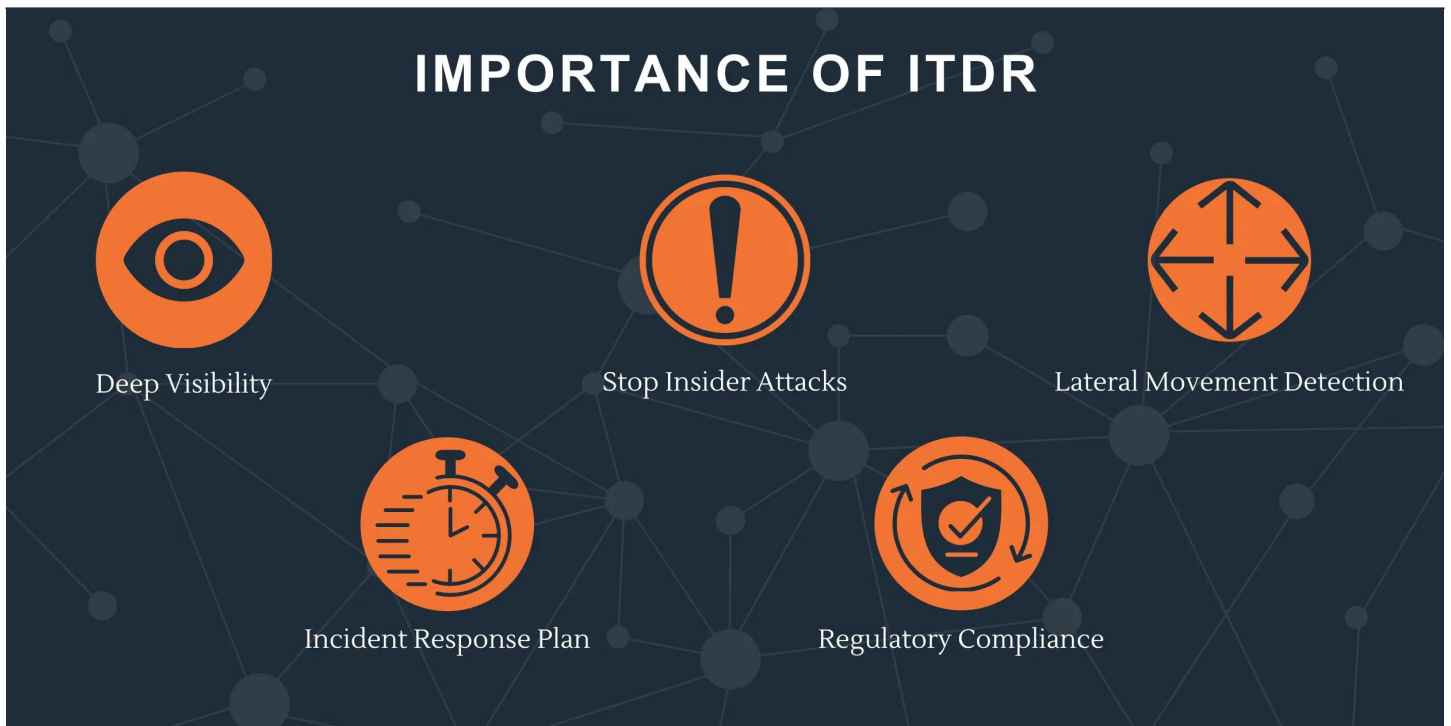
First, let's understand the importance of Identity Threat Detection and Response.

Importance of Identity Threat Detection and Response (ITDR)

According to research funded by the [Identity Defined Security Alliance \(IDSA\)](#), 79% of participants reported experiencing an identity-related breach in the previous two years.

As cyber threats evolve in complexity and frequency, the need for robust security measures to [protect sensitive data](#) and information has become critical. ITDR plays a crucial role in safeguarding organizational assets by focusing on the detection, investigation, and mitigation of identity-related threats.

IMPORTANCE OF ITDR



Here are some ways in which an ITDR system strengthens your enterprises defense set up:

- **Deep visibility into identity risks:** Continuous monitoring by ITDR helps proactively spot anomalous behaviors like unusual login attempts, access to unauthorized systems, or sudden spikes in activity. Detecting privilege escalation is crucial for identifying identity-based threats, as it can indicate malicious activity. This gives enterprises a higher chance of preventing the attack before it hits you.
- **Keeping an eye out for insider attacks:** Unsatisfied employees with access to company resources can often become unprecedented threat actors. ITDR can help uncover insider threats by flagging unusual access patterns from privileged accounts. With this early-stage information, security officials can revoke access and curb the threat.
- **Lateral Movement Detection:** In organizations which use [active directory](#), it's very easy for attackers to move from one account to another. ITDR can help avert attacks by having control over user permissions and accessing the potential pathway of the cyber attacker.
- **Strong Incident Response Plan:** In case of a breach, an ITDR can help reduce the detection and response time. This is possible because it can point out exactly which identity is compromised and quickly pull back access.
- **Maintaining Regulatory Compliance:** Among increasing regulatory compliance asking to [protect personal identifiable information \(PII\)](#), ITDR systems can help by taking a proactive approach for user access management.

Breaking down the ITDR Security Framework & Procedures

As we mentioned before, Identity threat detection & response (ITDR) is a security framework which often includes a bunch of procedures to help secure identity related threats

Here is a deeper look at that:

Identity and Access Management (IAM)

This is the process of centralizing and controlling user authentication and authorization process. This procedure often includes [multi-factor authentication \(MFA\)](#) to protect individual identities.

Threat Intelligence & Contextual Information

ITDR vendors must seek to integrate [threat intelligence feeds](#) to keep customers informed about emerging threats. Contextual information refers to tracking the source of potential identity threats.

Risk Assessment & Anomaly Behavior Detection

An ITDR solution must continuously update behavior models to adapt to evolving threat landscapes. Besides, not every anomaly poses a direct risk. Risk rating methods that rank warnings according to their possible impact and degree of severity should be included in the ITDR framework. This aids security teams in prioritizing the most important problems.

Response Automation & Mitigation

ITDR platforms and solutions have the ability to implement identity-specific incident response workflows to quickly isolate and contain compromised accounts. These platforms can swiftly isolate and contain compromised credentials to prevent further unauthorized access, thereby mitigating potential security risks.

Types of Identity Vulnerabilities

ITDR solutions can help mitigate identity vulnerabilities, which can be broadly categorized into three main types: unmanaged identities, misconfigured identities, and exposed identities. Each type presents unique risks and challenges that organizations must address to ensure robust identity protection.

Unmanaged Identities

These are user identities that are not properly managed or monitored, making them susceptible to data breaches. Unmanaged identities can include accounts that are not properly provisioned, deprovisioned, or updated. For example, former employees' accounts that are still active or service accounts that are not regularly reviewed can become easy targets for cybercriminals.

Misconfigured Identities

Misconfigured identities refer to accounts that are not properly configured, leaving them vulnerable to identity compromise. This can include user identities with excessive privileges, weak passwords, or inadequate access controls. For instance, an account with administrative privileges that does not require multi-factor authentication (MFA) can be a significant security risk.

Exposed Identities

These are identities that are exposed to the internet or other external networks, making them

vulnerable to attacks. Exposed identities can include accounts used for remote access, cloud services, or other external applications. For example, an identity used to access a cloud environment without proper security controls can be exploited by attackers to gain unauthorized access to sensitive data.

By understanding and addressing these types of identity vulnerabilities, organizations can strengthen their identity security policies and strategies and reduce the risk of identity related risks.

Implementing ITDR in Your Organization

Implementing Identity Threat Detection and Response (ITDR) in your organization requires a strategic approach that involves several key steps. Here are some best practices to consider:

- **Identify Your Identity Attack Surface**

Begin by understanding the identity landscape that exists within your organization and the potential attack vectors that attackers may use to exploit them. This includes user accounts, service accounts, and privileged accounts.

- **Assess Your Current Security Controls**

Evaluate your current security controls and identify gaps in your identity protection strategy. This involves reviewing your identity and access management (IAM) practices, multi-factor authentication (MFA) implementation, and privileged access management (PAM) controls.

- **Implement ITDR Solutions**

Deploy ITDR solutions that can detect and respond to identity-based threats in real-time. These solutions should provide continuous monitoring, [threat intelligence](#), and automated response capabilities to quickly identify and mitigate identity threats.

- **Monitor and Analyze User Activity**

Continuously monitor and analyze user activity to detect and respond to identity-based threats. This includes analyzing access management logs, network traffic, and user behavior to identify anomalies and potential security incidents.

- **Continuously Update and Refine Your ITDR Strategy**

Stay ahead of emerging threats and vulnerabilities by continuously updating and refining your ITDR strategy. This involves regularly reviewing and updating your security controls, threat intelligence feeds, and incident response plans.

By following these best practices, organizations can effectively implement ITDR and enhance their identity protection strategies, reducing the risk of identity-based attacks.

The Future of ITDR

The future of Identity Threat Detection and Response (ITDR) is rapidly evolving, with new technologies and innovations emerging to address the growing threat of identity-based attacks.

Here are some trends to watch:

1. **Machine Learning:** ML will continue to play a critical role in ITDR, enabling organizations to detect and respond to identity-based threats in real-time. These technologies can analyze vast amounts of data, [identify patterns](#), and predict potential threats, enhancing the effectiveness of ITDR solutions.
2. **Cloud-Based ITDR Solutions:** Cloud-based ITDR solutions will become increasingly popular, providing organizations with scalable and flexible ITDR capabilities. These solutions can be easily deployed and managed, offering real-time threat detection and response without the need for significant investment in infrastructure.
3. **Identity Protection as a Service:** Identity protection as a service will become a growing trend, providing organizations with comprehensive ITDR capabilities without the need for significant investment in infrastructure and personnel. This approach allows organizations to leverage the expertise of ITDR providers and focus on their core business activities.
4. **Integration with Other Security Solutions:** ITDR solutions will increasingly integrate with other security solutions, such as EDR, Active Directory Security and Security Information and Event Management (SIEM) systems, to provide a comprehensive security posture. This integration enables organizations to correlate data from multiple sources, enhancing their ability to detect and respond to complex threats.

Connection between Active Directory and ITDR

As the house of user accounts, permissions, and access controls, Active Directory is the prime target for cyber attackers. Identity Threat Detection & Response solutions can integrate with Active Directory to gain valuable insights into user activity within this core system. With the use of advanced analytics and machine learning, this integration enables the real-time identification of abnormalities, including unwanted access attempts and privilege escalations.

Detecting privilege escalation is crucial for identifying identity-based threats, alongside other threats like account takeovers and data exfiltration. ITDR improves your overall security posture and gives you the ability to recognize and counter [threats that are directed toward your Active Directory infrastructure](#) by continuously monitoring this single identity hub. So, although an ITDR and an AD security tool have their own capabilities and use cases, in simplified terms an AD security solution is a version of ITDR specifically designed to protect from identity threats within an active directory only.

Fidelis Active Directory Intercept™

If you are looking for a solution that spreads a wider net than ITDR may be the way to go, but for a micro picture of your Active Directory, [Fidelis Active Directory Intercept™](#) is the way to go.

It is the only solution which combines AD-aware [network detection and response \(NDR\) tool](#) and integrated Active Directory [deception technology solution](#) with foundational AD log and event monitoring to not just identify Microsoft Active Directory threats – but to respond swiftly.

Here are some benefits of Fidelis Active Directory Intercept™ solution:

- It enables you to catch risks coming from multiple sources.
- It provides complete and contextual visibility into your enterprise resources.
- It enables security teams to seamlessly catch configuration issues caused due to ongoing changes in Microsoft Active Directory

Enhanced AD Threat Detection and Defense

Deploy Fidelis Active Directory Intercept™ and give your enterprise the power to see more, detect faster, defend better and respond timely.

[Talk to an expert](#)

Conclusion

In summary, ITDR is a critical component of a comprehensive cybersecurity strategy, and its importance will only continue to grow in the future. By understanding the types of identity vulnerabilities, implementing ITDR solutions, and following best practices, organizations can protect themselves against identity-based attacks and stay ahead of emerging threats and vulnerabilities.